Problem Statement 9

Automation of Complex Decision-Making using AI

BACKGROUND

Modern systems generate large volumes of structured and unstructured data, making decision-making complex and time-consuming. Many real-world problems involve multiple parameters and require expert judgment. AI-driven decision automation systems can help model such complexity and enable faster, data-driven decisions.

OBJECTIVE

Identify and design use cases for automating complex decision-making

Develop a data-driven model to support and automate such decisions

EXPECTED OUTCOMES

Well-defined use cases involving complex decision processes

Decision model capturing parameters, relationships, and outcomes

Framework for automated scoring and recommendation

DATASET REQUIRED

DOMAIN DATA

Depends on selected use case (e.g., healthcare, finance, governance)

PARAMETERS

Structured and unstructured data relevant to decision variables

EXPERT INPUT

Domain knowledge for defining relationships and rules

SUGGESTED TOOLS/TECHNOLOGIES

PROGRAMMING

Python

AI/ML

Bayesian Networks, decision trees, probabilistic models

PLATFORMS

No-code/low-code tools, data processing frameworks

EXPECTED SOLUTION / STEPS

Identify real-world use cases requiring complex decisions

Define parameters, variables, and possible outcomes

Model relationships using decision trees or Bayesian networks

Build inference/scoring system for decision automation

Validate model using data and expert input

Present solution with clear explanation and use-case impact

EVALUATION PARAMETERS

Complexity and relevance of chosen use case

Quality of decision modeling and parameter definition

Effectiveness of automation in solving real-world problems

Clarity of approach and presentation